Service levels

NetApp Keystone

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Service levels

Use the methods listed in the following table to retrieve, create, modify, and delete service levels.

HTTP Method	Path	Description
`GET`	`/v2.1/servicelevels`	Retrieve all service levels.
`GET`	`/v2.1/servicelevels/{id}`	Retrieve a service level by ID.
`POST`	`/v2.1/servicelevels`	Create a new service level.
`PUT`	`/v2.1/servicelevels/{id}`	Modify service level details by ID.
`DELETE`	`/v2.1/servicelevels/{id}`	Delete a service level.

Service level attributes

The following table lists the service level attributes.

Attribute	Type	Description
`id`	String	The unique identifier for the service level.
`name`	String	The service level name.
`description`	String	The description of the service level.
`policy_name`	String	Service level quality of service (QoS) name. Allowed values: nse_value, nse_standard, nse_performance, and nse_extreme.
`available`	Boolean	Indicates whether the service level is available for use.
`grandfathered`	Boolean	Indicates whether the service level has been retired.
`peak_iops_per_tb`	Integer	The maximum possible IOPS per TiB.
`expected_iops_per_tb`	Integer	The minimum expected IOPS per TiB.
`absolute_min_iops`	Integer	The absolute minimum IOPS which is used as an override when the expected IOPS is less than this value.
`peak_iops_allocation`	String	The peak IOPS allocation. Allowed values: allocated_space and used_space.
`io_block_size_kb`	Integer	The I/O block size (KiB).

Attribute	Type	Description
`min_size_gb`	Integer	The minimum size (GiB).
`max_size_gb`	Integer	The maximum size (GiB).
`max_peak_iops`	Integer	The maximum peak IOPS for the service level.
`max_expected_iops`	Integer	The maximum expected IOPS.
`autogrow_max_percent`	Integer	The autogrow maximum percent.
`ontap_aggregates`	–	The list of ONTAP aggregates. An ONTAP aggregate consists of: • `cluster_UIID: ONTAP cluster UUID` • `aggr_name: ONTAP aggregate name` • `aggr_uuid: ONTAP aggregate UUID` • `node_name: ONTAP cluster node name` For example: `"ontap_aggregates": [ { "cluster_uuid": "3fa85f64-5717-4562-b3fc-2c963f66afa6", "aggr_name": "string", "aggr_uuid": "3fa85f64-5717-4562-b3fc-2c963f66afa6", "node_name": "node01" } ]`

Attribute	Type	Description
`primary_volume_defaults`	–	• snapshot_auto_delete_target_free_space (integer): This option specifies the free space percentage at which the automatic deletion of Snapshot copies must stop. • auto_size_mode (string): The autosize mode for the volume. Allowed values: off, grow, grow_shrink For example: `"primary_volume_defaults": { "snapshot_auto_delete_target_free_space": 3, "auto_size_mode": "grow_shrink"`

Retrieve service levels

Use the method listed in the following table to retrieve all service levels.

HTTP Method	Path	Description	Parameters
`GET`	`/v2.1/servicelevels`	Retrieve all service levels.	None

Required request body attributes: `none`

Request body example:

```
none
```

Response body example:

```
{
  "status": {
    "user_message": "Okay. Returned 3 records.",
    "verbose_message": "",
    "code": 200
  },
  "result": {
```

```
    "total_records": 3,
    "records": [
      {
        "name": "standard",
        "description": "Best suited for general purpose workloads",
        "slo": "1000IOPS/TB",
        "min_size": 137438953472,
        "io_block_size_kb": 32,
        "min_size_gb": 10,
        "max_size_gb": 40960,
        "min_iops": 100,
        "peak_iops_per_tb": 1000,
        "expected_iops_per_tb": 700,
        "max_peak_iops": 40000,
        "max_expected_iops": 28000,
        "max_peak_throughput": 1250,
        "max_expected_throughput": 875
      },
      {
        "name": "extreme",
        "description": "Best suited for performance-critical workloads",
        "slo": "12000IOPS/TB",
        "min_size": 91625968981,
        "io_block_size_kb": 32,
        "min_size_gb": 10,
        "max_size_gb": 10240,
        "min_iops": 500,
        "peak_iops_per_tb": 12000,
        "expected_iops_per_tb": 8000,
        "max_peak_iops": 120000,
        "max_expected_iops": 60000,
        "max_peak_throughput": 3750,
        "max_expected_throughput": 1875
      },
      {
        "name": "premium",
        "description": "Best suited for databases and high performance
workloads",
        "slo": "4000IOPS/TB",
        "min_size": 137438953472,
        "io_block_size_kb": 32,
        "min_size_gb": 10,
        "max_size_gb": 10240,
        "min_iops": 300,
        "peak_iops_per_tb": 4000,
        "expected_iops_per_tb": 3000,
```

```
        "max_peak_iops": 40000,
        "max_expected_iops": 30000,
        "max_peak_throughput": 1250,
        "max_expected_throughput": 937
      }
    ]
  }
}
```

Retrieve service levels by name

Use the method listed in the following table to retrieve service levels by name.

HTTP Method	Path	Description	Parameters
`GET`	`/v2.1/servicelevels/{name}`	Retrieve a service level by name.	`name (string)`: The name of the service level.

Required request body attributes: `none`

Request body example:

```
none
```

Response body example:

```
{
  "status": {
    "user_message": "Okay. Returned 1 record.",
    "verbose_message": "",
    "code": 200
  },
  "result": {
    "returned_records": 1,
    "records": [
      {
        "name": "premium",
        "description": "Best suited for databases and high performance
workloads",
        "slo": "4096IOPS/TB",
        "min_size": 137438953472,
        "io_block_size_kb": 32,
        "min_size_gb": 10,
        "max_size_gb": 10240,
        "min_iops": 300,
        "peak_iops_per_tb": 4096,
        "expected_iops_per_tb": 3000,
        "max_peak_iops": 40000,
        "max_expected_iops": 30000,
        "max_peak_throughput": 1250,
        "max_expected_throughput": 937
      }
    ]
  }
}
```


Create a service level

Use the method listed in the following table to create a service level.

HTTP Method	Path	Description	Parameters
`POST`	`/v2.1/servicelevels`	Create a service level.	None

Required request body attributes: `name`, `policy_name`

Request body example:

```
{
  "name": "MyServiceLevelName",
  "description": "My new service level description",
  "policy_name": "nse_value",
  "available": true,
  "grandfathered": false,
  "peak_iops_per_tb": 1000,
  "expected_iops_per_tb": 700,
  "absolute_min_iops": 100,
  "peak_iops_allocation": "allocated_space",
  "io_block_size_kb": 32,
  "min_size_gb": 10,
  "max_size_gb": 40960,
  "max_peak_iops": 20000,
  "max_expected_iops": 5000,
  "autogrow_max_percent": 3,
  "ontap_aggregates": [
    {
      "cluster_uuid": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
      "aggr_name": "string",
      "aggr_uuid": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
      "node_name": "node01"
    }
  ],
  "primary_volume_defaults": {
    "snapshot_auto_delete_target_free_space": 3,
    "auto_size_mode": "grow_shrink"
  }
}
```


Response body example:

```
{
  "status": {
    "user_message": "Okay. New resource created.",
    "verbose_message": "",
    "code": 201
  },
  "result": {
    "total_records": 1,
    "records": [
      {
        "name": "MyServiceLevelName",
        "description": "My new service level description",
        "slo": "1000IOPS/TB",
        "min_size": 0,
        "io_block_size_kb": 32,
        "min_size_gb": 10,
        "max_size_gb": 40960,
        "min_iops": 100,
        "peak_iops_per_tb": 1000,
        "expected_iops_per_tb": 700,
        "max_peak_iops": 20000,
        "max_expected_iops": 5000,
        "max_peak_throughput": 625,
        "max_expected_throughput": 156
      }
    ]
  }
}
```


Modify a service level

Use the method listed in the following table to modify a service level.

HTTP Method	Path	Description	Parameters
`PUT`	`/v2.1/servicelevels/{name}`	Modify the details of a service level.	`name (string)`: The name of the service level.

Required request body attributes: `none`

Request body example:

```
{
  "name": "MyNewServiceLevelName",
  "description": "Service level description",
  "policy_name": "nse_value",
  "available": false,
  "grandfathered": false,
  "peak_iops_per_tb": 1000,
  "expected_iops_per_tb": 700,
  "absolute_min_iops": 100,
  "peak_iops_allocation": "allocated_space",
  "io_block_size_kb": 32,
  "min_size_gb": 10,
  "max_size_gb": 40960,
  "max_peak_iops": 20000,
  "max_expected_iops": 5000,
  "autogrow_max_percent": 3,
  "ontap_aggregates": [
    {
      "cluster_uuid": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
      "aggr_name": "string",
      "aggr_uuid": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
      "node_name": "node01"
    }
  ],
  "primary_volume_defaults": {
    "snapshot_auto_delete_target_free_space": 3,
    "auto_size_mode": "grow_shrink"
  }
}
```

Response body example:

```
TBA
```

Delete service level by ID

Use the method listed in the following table to delete a service level by ID.

HTTP Method	Path	Description	Parameters
`DELETE`	`/v2.1/servicelevels/{name}`	Delete the service level identified by ID.	`name (string)`: The name of the service level.

Request body example:

```
none
```

Response body example:

```
No content for succesful delete
```

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